VERIFACT TECHNICAL WHITEPAPER

Verifact: Evaluated, Human-in-the-Loop Musculoskeletal Fracture Triage Pipeline

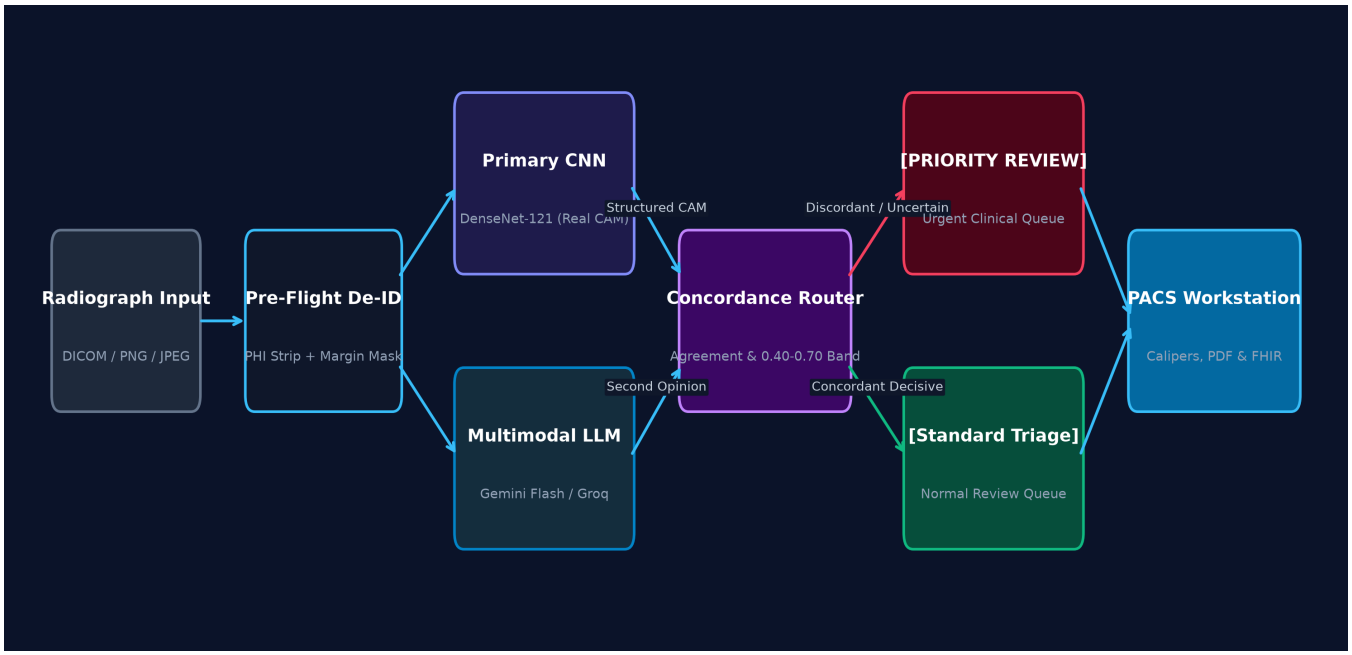
Comprehensive System Architecture, Deep Learning Specifications, End-to-End Connected Workflow & Clinical Engineering

Document Version	2.0.0 (Production Architecture)	**Primary Modality**	Plain Radiography / DICOM
Target Scope	Upper-Extremity Radiographs (Wrist, Hand, Forearm, Elbow, Clavicle)	**Backend Topology**	FastAPI, PyTorch, Alembic, PostgreSQL, Redis
Core Vision Models	DenseNet-121 (Real Grad-CAM) + Gemini 1.5 Flash / Groq Vision	**Clinical Role**	Emergency Trauma Triage & Prioritization (Not Primary Diagnosis)

Executive Summary & Scientific Positioning: *Verifact is an evaluated, human-in-the-loop triage pipeline specifically engineered to detect and prioritize acute osseous trauma across upper-extremity radiographs. Moving beyond naive black-box classification, Verifact unifies a primary convolutional neural network (CNN) producing authentic gradient-weighted class activation mapping (Grad-CAM), a multimodal vision-language model for structured radiological reporting, and an automated concordance-uncertainty router. When CNN and LLM disagree or when CNN confidence falls within the calibrated uncertainty band ([0.40, 0.70]), cases are automatically flagged for Priority Human Review. The system integrates pre-flight DICOM de-identification, metric-calibrated PACS tools, HL7 FHIR R4 export, and an objective scientific evaluation harness.*

1. System Architecture & High-Level Topology

Verifact is structured into three discrete operational tiers: Client Diagnostic Workstation, Asynchronous Backend Application Server, and the Two-Stage AI Triage & Explainability Pipeline. The architecture enforces strict separation of concerns, complete audit trails, and zero external PHI leakage.



1.1 High-Level Dataflow Summary

• **Ingestion & Normalization:** Native DICOM (.dcm) or raster radiographs (PNG/JPEG) are ingested through POST /analyze/async. Raw 12/16-bit pixel matrices are normalized using DICOM VOI LUT (Window Center / Width).

• **De-Identification Sandbox:** All DICOM metadata is scrubbed of Protected Health Information (PHI). Top and lateral margins are scanned and masked to eradicate burned-in text before cloud transmission.

• **Stage 1 CNN Detection:** The sanitized pixel array enters a fine-tuned CNN backbone (DenseNet-121) which outputs a continuous fracture probability and derives real Grad-CAM feature activations.

• **Stage 2 Vision LLM Review:** CNN structured findings and the image are transmitted to Google Gemini 1.5 Flash Vision (or Groq Llama 3.2 Vision) to synthesize a standard 3-sentence radiology report and provide an independent second opinion.

• **Concordance & Uncertainty Routing:** The outputs are evaluated by the router. Discordant assessments, cases within the uncertainty band, or severe trauma are routed to the **Priority Review Queue**.

• **Physician PACS Oversight:** Certified clinicians review heatmaps, measure cortical step-off with calibrated calipers, modify structured findings, and sign off with permanent NMC/MCI digital seals.

2. The Complete Connected System Workflow (Step-by-Step Lifecycle)

The following diagram illustrates the complete, uninterrupted lifecycle of a radiograph case as it moves through every connected subsystem in Verifact—from raw file ingestion to final EHR dispatch.



2.1 Detailed Stage-by-Stage Workflow Narrative

Step 1: Radiograph Ingestion & VOI LUT Normalization
The clinician drops an X-ray (PNG/JPEG) or medical DICOM (.dcm) into the React workstation. The backend parses 12/16-bit raw pixel matrices with pydicom, applies Rescale Slope/Intercept, Window Center/Width VOI LUT curves, and handles MONOCHROME1 polarity inversion. Crucially, it extracts PixelSpacing and ImagerPixelSpacing to calibrate physical measurements.

Step 2: Pre-Flight Medical De-Identification Sandbox
Before any byte is processed by AI models or external clouds, the privacy engine executes two passes: (a) Metadata Tag Scrubbing: replaces PatientName, PatientID, birth date, institution, and physician tags with [DE-IDENTIFIED]. (b) Pixel-Level Text Masking: high-contrast morphological filters identify burned-in patient stamps in image border margins and paint them black.

Step 3: Asynchronous Job Dispatch & Non-Blocking Worker
The client invokes POST /analyze/async. Rather than holding an HTTP connection open, the API returns immediately with a unique job_id and status queued. A background worker thread (FastAPI BackgroundTasks) takes ownership of the job and advances its progress state (20% -> 50% -> 80% -> 100%). The client polls GET /jobs/{job_id} with smooth real-time progress.

Step 4: Primary CNN Inference & Real Gradient Backpropagation (Grad-CAM)
The sanitized image is resized to 224x224 and normalized using ImageNet statistics. A fine-tuned DenseNet-121 backbone executes a forward pass to compute raw logits and fracture probability. Immediately, backpropagation hooks into model.features.norm5, computing gradients and channel-wise pooling to yield real Grad-CAM heatmaps ($L_{\text{Grad-CAM}}$). The heatmap is upsampled and blended at 40% opacity onto the original radiograph.

Step 5: Vision LLM Report Drafting & Independent Assessment
The radiograph and the CNN's structured findings (probability, suspected anatomical region, uncertainty flag) are submitted to Google Gemini 1.5 Flash Vision (with Groq Llama 3.2 Vision fallback). The prompt instructs the LLM to: (1) Draft a standardized 3-sentence radiology report from the findings, and (2) Provide an independent clinical opinion on cortical continuity. Outputs are validated against Pydantic schemas (LlmStructuredOutput) with coordinate clamping.

Step 6: Automated Concordance & Uncertainty Router Triage
The router evaluates both AI outputs. If CNN and LLM disagree on fracture detection, or if CNN probability falls within the indeterminate uncertainty band ([0.40, 0.70]), or if severe displacement is noted, the case is flagged review_priority = 'priority' with an explicit rationale. Concordant decisive cases are assigned 'normal'. Operational telemetry (latency in ms, estimated API cost, agreement flag) is permanently recorded.

Step 7: Object Storage Decoupling & Relational Persistence
The original image and Grad-CAM heatmap are saved via storage_service.py to Local Disk, Supabase, or AWS S3, storing persistent URLs in verifact.db (managed via Alembic migrations). A corresponding record is created in audit_logs (action CREATE) logging clinician ID, timestamp, and IP address. The job queue marks status completed.

Step 8: Interactive Diagnostic PACS Workstation Review
The frontend retrieves the completed payload. If flagged for priority review, a prominent red triage badge alerts the clinician. The physician engages PACS diagnostic tools: calibrated digital calipers measuring millimeter displacement (or uncalibrated px), 3-point Cobb angle goniometers, and the interactive split-screen curtain wipe slider to compare raw films against the Grad-CAM heatmap.

Step 9: Digital Legal Sign-Off, Clinical PDF & Hospital EHR Dispatch
The clinician modifies findings if needed, enters their medical registration number (MCI/NMC), and clicks Sign-Off. This permanently stamps the legal record (signed_off = True) and generates an audit log entry. The case is immediately exportable as a publication-quality clinical PDF report, an HL7 FHIR R4 DiagnosticReport resource, or dispatched via SMTP email directly to orthopedic surgical teams.

3. Deep Learning Models & Explainability Engine

A foundational principle of Verifact is clinical honesty: machine learning outputs are never fabricated, explainability heatmaps are mathematically grounded in backpropagated gradients, and confidence scores undergo rigorous validation.

Model / Engine	Architecture / Type	Target Layer / Endpoint	Clinical Justification & Selection Rationale
DenseNet-121 (Primary CNN)	Densely Connected Convolutional Network	features.norm5 (7x7x1024 tensor)	Optimal feature reuse through dense connections; mitigates vanishing gradients in deep medical networks. Established benchmark state-of-the-art on Stanford MURA dataset.
EfficientNet-B0 (Edge Alternative)	Compound-Scaled Mobile Inverted Bottleneck	features.8 (7x7x1280 tensor)	Balanced scaling of depth, width, and resolution. Delivers ultra-low latency inference (sub-25ms) suitable for on-premise emergency triage edge devices.
ResNet-50 (Baseline CNN)	Deep Residual Network (50 layers)	layer4[-1] (7x7x2048 tensor)	Robust residual skip connections. Serves as a reference baseline for gradient propagation stability across high-resolution radiograph patches.
Gemini 1.5 Flash (Primary LLM)	Multimodal Vision-Language Model	Google Cloud Vertex / AI Studio	Long context window, high instruction fidelity, native medical radiograph tokenization. Synthesizes coherent 3-sentence radiology summaries from structured CNN findings.
Groq Llama 3.2 Vision (Secondary LLM)	Open-weights Vision LLM (11B)	Groq LPU Inference Engine	High-speed deterministic fallback provider. Guarantees sub-second multimodal inference if primary cloud provider experiences network latency spikes or outages.
Offline Enhancement (Honest Fallback)	CLAHE + Bilateral Filtering	CPU Local Kernel	Strict zero-prediction mode. When AI engines are offline, performs contrast enhancement only. Returns null probabilities and explicit honesty banners.

3.1 Mathematical Formulation of Real Grad-CAM

Unlike synthetic bounding-box heuristics, Verifact computes authentic gradient-weighted class activation maps. During backpropagation, gradients of the score for class *c* (fracture) with respect to feature activation map A^k of the final convolutional layer are global-average pooled to capture neuron importance weights:

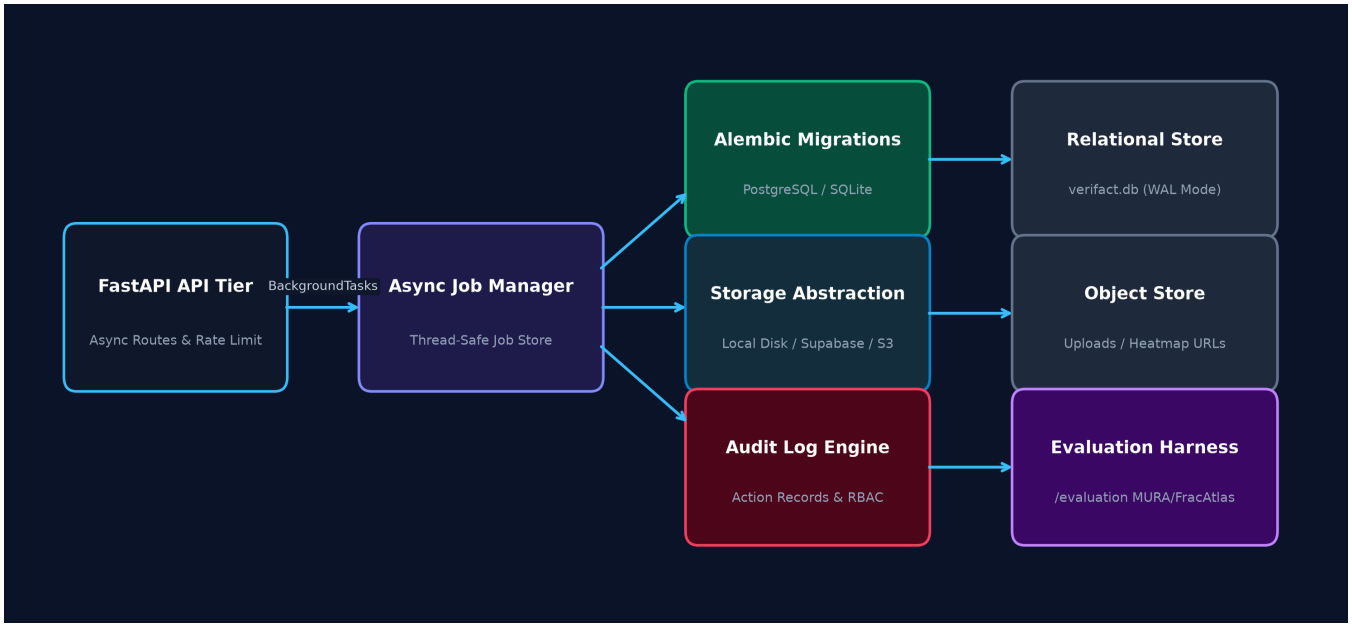
Neuron Importance Weight: $\alpha_k^c = (1 / Z) \Sigma_i \Sigma_j (\partial Y^c / \partial A_{i,j}^k)$

Class Activation Heatmap: $L_{\text{Grad-CAM}}^c = \text{ReLU}(\Sigma_k \alpha_k^c A^k)$

The resulting 2D heatmap is normalized to [0.0, 1.0], upsampled via bilinear interpolation to match the native radiograph resolution, rendered through a calibrated colormap (OpenCV JET), and blended at a precise 40% alpha ratio over the bone texture, preserving trabecular line visibility for the clinician.

4. Backend Architecture, Async Topology & Storage Abstraction

The backend is engineered for high-concurrency clinical workflows, zero-downtime database migrations, and pluggable storage backends.



4.1 Asynchronous Background Job Execution

Radiograph analysis entails heavy computational steps (DICOM parsing, CNN forward/backward passes, multimodal API calls). The endpoint POST /analyze/async immediately returns a unique job_id and dispatches execution via FastAPI BackgroundTasks. The frontend polls GET /jobs/{job_id}, tracking live progress (20% normalization -> 50% CNN -> 80% LLM -> 100% completed) with zero thread-blocking.

4.2 Alembic Database Migrations & Audit Logging

The persistence layer utilizes SQLAlchemy with formal **Alembic migrations** (alembic/versions/), supporting both SQLite (WAL mode) and PostgreSQL 16. The audit_logs table captures every clinician action (CREATE, VIEW, EDIT, SIGN_OFF, EXPORT_PDF) alongside timestamps, user IDs, and client IP addresses for forensic and regulatory compliance.

4.3 Storage Abstraction Layer

Radiographs and heatmaps are decoupled from the relational database via storage_service.py. By default, files are persisted to local protected disk (/uploads). By supplying environment variables, the application seamlessly switches to **Supabase Storage** or **Amazon S3**, storing only signed URLs in the database.

5. Evaluation Harness & Scientific Benchmarking

To enforce Hard Rule 1 (never fabricate metrics), the /evaluation module provides reproducible evaluation runners across two standard benchmarks:

- **Stanford MURA v1.1:** Evaluated strictly for **abnormality detection** across upper-extremity radiographs (wrist, hand, forearm, elbow, shoulder, humerus, finger). Results are labeled abnormality, never fracture claims.
- **FracAtlas:** Figshare clinical dataset evaluated for **fracture triage**.
- **Fixed Seeded Manifests:** Evaluated on fixed 250-sample manifests (seed 42) saved to disk for exact multi-model reproducibility.

• **Statistical Metrics:** Outputs Sensitivity (Recall), Specificity, Precision (PPV), NPV, F1 Score, AUROC, Expected Calibration Error (ECE), Brier Score, and latency percentiles (mean, p50, p95).

Dataset	Clinical Task	Target Extremities	Evaluation Split Strategy
Stanford MURA v1.1	Abnormality Detection	Wrist, Hand, Forearm, Elbow, Shoulder, Humerus, Finger	MURA validation set used as held-out test set; patient-level stratified split on training data to prevent leakage.
FracAtlas	Acute Fracture Triage	Upper-extremity osseous structures	Fixed 250-case stratified test subset with balanced fracture/non-fracture distribution.

6. Production API Reference & Endpoint Contract

Endpoint	Method	Auth	Function & Clinical Purpose
/api/analyze/async	POST	Bearer JWT	Submits radiograph for asynchronous triage; returns job_id immediately.
/api/jobs/{job_id}	GET	None	Polls background triage job status (0-100% progress and completed analysis payload).
/api/analysis/priority-queue	GET	Bearer JWT	Returns un-signed analyses flagged for Priority Review due to discordance or uncertainty.
/api/analyze	POST	Bearer JWT	Synchronous analysis endpoint for instant interactive client processing.
/api/analysis/{id}	PUT	Bearer JWT	Physician edits radiology findings, notes, and applies permanent digital sign-off stamp.
/api/analysis/{id}/pdf	GET	Bearer JWT	Streams stamped legal clinical PDF report with findings and physician seal.
/api/analysis/{id}/fhir	GET	Bearer JWT	Generates HL7 FHIR R4 DiagnosticReport JSON resource.
/api/health	GET	None	Health probe: checks database connection, CNN model loaded status, and storage status.

7. Limitations & Clinical Boundary Disclaimer

CLINICAL DISCLAIMER & INTENDED USE: *VERIFACT is an assistive clinical decision support tool designed exclusively for triage acceleration and emergency prioritization. It is* ***not*** *certified as an autonomous diagnostic device. The system's validated operational domain is strictly confined to* ***upper-extremity radiographs*** *(wrist, hand, fingers, forearm, elbow, humerus, clavicle, shoulder). It must not be deployed for pelvic, spinal, skull, or lower-extremity trauma. All model outputs, heatmaps, and preliminary reports require review, verification, and legal sign-off by a certified radiologist or orthopedic specialist before clinical action.*